

1 Functional Requirements Specification (FRS)

1.1 Core Actors

1.2 Functional Requirements

1.2.1 User Interaction (Text and Voice)

- **FR1:** The system shall allow users to send **text messages** through the chat widget or Meta channels.
 - **FR2:** The system shall support **voice input** (speech-to-text) and **voice output** (text-to-speech) via Whisper and TTS APIs (future).
 - **FR3:** The system shall store each message or voice record in **ChatMessage** or **VoiceInteraction** with metadata such as AI model used, detected emotion, timestamp, content or transcription text, voice URL, and sender type.
 - **FR4:** The system shall detect user **emotion** from the text or voice tone using an emotion detection layer.
 - **FR5:** The chatbot shall **adapt its tone or response style** based on the detected emotion and business tone configuration.
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1.2.2 Widget & Meta Integration

- **FR6:** The system shall provide an **embeddable chat widget** using a unique **api_key** from **BusinessAuth**.
- **FR7:** The widget shall support **real-time messaging**, **typing indicators**, and **session persistence**.
- **FR8:** The system shall verify the **api_key** before initializing any chat session.
- **FR9:** The system shall integrate with **WhatsApp**, **Messenger**, and **Instagram APIs** (planned feature) for businesses to connect their customer channels.
- **FR10:** The widget and integrations shall automatically switch to **offline mode** outside business hours (**operating_hour_start** / **operating_hour_end**).

1.2.3 Knowledge Base and AI Response

- **FR11:** The business admin shall be able to upload or input **FAQs, text, or files** into the **KnowledgeBase** (text or file-based).
 - **FR12:** The AI pipeline shall generate **embedding vectors** for each entry using AI frameworks (e.g., LangChain) and store them in the **embedding_vector** field.
 - **FR13:** When a user message is received, the system shall perform **contextual retrieval (RAG)** to fetch top-matching knowledge chunks.
 - **FR14:** The AI controller layer shall choose the most confident model output between **NileChat, Gemini, or Qwen** based on similarity threshold.
 - **FR15:** The system shall format AI responses based on business rules from **BusinessSetting**, consistently using a **response formatting layer**.
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1.2.4 Business Registration and Authentication

- **FR16:** Businesses can register an account providing **name, email, and domain_url**.
 - **FR17:** A default **BusinessSetting** and active **BusinessAuth** record shall be automatically generated upon registration.
 - **FR18:** Each business can manage multiple **BusinessAdmin** accounts with roles (**owner, staff**).
 - **FR19:** Authentication shall be done through secure login using email and password.
 - **FR20:** The system shall allow owners to **generate, rotate, or deactivate API keys** without deleting the business record.
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1.2.5 Dashboard Management

- **FR21:** The business admin shall access an **admin dashboard** via web (React.js).
 - **FR22:** The dashboard shall allow management of:
 - Knowledge Base entries (CRUD)
 - Bot tone style and response mode (auto/manual)
 - Operating hours
 - Viewing chat history and feedback
 - **FR23:** The dashboard shall show analytics reports such as total chats, average response time, and user satisfaction (from **AnalyticsReport**).
 - **FR24:** The dashboard shall include a **manual reply mode**, allowing admins to respond directly to ongoing chats if AI auto-reply is disabled.
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1.2.6 Chat Session Management

- **FR25:** When a user starts chatting, the system shall create a new **ChatSession** linked to the **EndUser** and **Business**.
 - **FR26:** All messages within the same chat shall link to that session ID for history tracking.
 - **FR27:** When a chat ends, the system shall mark the session as **closed** or **expired**.
 - **FR28:** The user shall be prompted to submit **feedback** (rating and comment), stored in the **Feedback** table.
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1.2.7 Analytics and Reporting

- **FR29:** The system shall automatically aggregate chat statistics and feedback to generate **AnalyticsReport** entries.
- **FR30:** Reports shall include total chats, average response time, and average user satisfaction.
- **FR31:** Reports shall be available for download or viewing in the dashboard.

- **FR32:** Analytics shall update periodically (e.g., daily or weekly background job).
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1.2.8 Voice Interaction (Future Expansion)

- **FR33:** The system shall store voice recordings and transcription text in the `VoiceInteraction` table.
 - **FR34:** The system shall detect emotion from voice tone and integrate results with AI response generation.
 - **FR35:** The chatbot shall be able to **reply with customized voice** based on user preference or platform type (e.g., WhatsApp voice message).
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1.2.9 AI Pipeline Orchestration

- **FR36:** The AI Orchestrator shall manage the workflow between input message → emotion detection → RAG → model selection → formatting → output.
 - **FR37:** The orchestrator shall fallback between models if the primary fails.
 - **FR38:** Each step of the pipeline shall log its operation (model used, latency, confidence score).
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1.2.10 System Administration

- **FR39:** The system shall maintain logs of authentication attempts, API usage, and integration events.
- **FR40:** The system shall support environment-based configurations for development, staging, and production.
- **FR41:** The system shall notify admins of integration errors (e.g., API key invalid, channel disconnected).

2 Non-Functional Requirements

2.1 Performance

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- **NFR-1:** Average chatbot response (text) shall be less than or equal to **5 seconds** for 90% of requests.
- **NFR-2:** Emotion detection latency shall not exceed **600 ms for text** , **1000ms for voice**, and RAG retrieval shall not exceed **1.5 s for text** , **3 s for voice**.
- **NFR-3:** Voice query processing (audio → text → reply) shall take less than 1 min
- **NFR-4:** Average database query latency shall not exceed **200 ms** under normal load.

2.2 Scalability

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- **NFR-5:** The architecture shall support **horizontal scaling** using Docker and (Railway or Coolify) .
- **NFR-6:** The queue system (Redis "caching") shall buffer up to **2k concurrent jobs** without failure.

2.3 Reliability & Availability

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- **NFR-7:** The system shall maintain an uptime of at least **95% per month**.
- **NFR-8:** Critical services shall automatically restart within **30 seconds** after failure.
- **NFR-9:** Database backups shall be taken **daily** and retained for **14 days**.
- **NFR-10:** Model fallback (NileChat → Gemini → Qwen) success rate shall be at least **95%**.

2.4 Security

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- **NFR-11:** All traffic shall use **HTTPS/TLS 1.3**.
- **NFR-12:** Passwords shall be hashed using **bcrypt (cost ≥ 12)**, and API keys hashed using **SHA-256**.
- **NFR-13:** JWT access tokens shall expire in **24 hours**, and refresh tokens shall be valid for **7 days**.
- **NFR-14:** Accounts shall be locked after **5 failed login attempts within 30 minutes**.
- **NFR-15:** Sensitive data shall be encrypted at rest using **AES-256**.

2.5 Usability

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- **NFR-16:** Chat widget shall load in less than or equal to **30 seconds**
- **NFR-17:** Dashboard shall be responsive for screens wider than **768 px**.
- **NFR-18:** System shall support both **Arabic (Egyptian dialect)** and **English**, with automatic RTL layout.
- **NFR-19:** At least **80% of users** shall be able to complete a chat without assistance

2.6 Maintainability & Testability

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- **NFR-20:** Codebase shall be organized into modular components: **api**, **ai_pipeline**, **db**, **dashboard**, and **integrations**.
- **NFR-21:** Unit-test coverage shall be at least **70%** for backend components.
- **NFR-22:** The CI/CD pipeline (GitHub Actions) shall complete build and deployment in less than **8 minutes**.
- **NFR-23:** All APIs shall include auto-generated documentation using **Swagger** or **ReDoc** and manual postman doc for testing.

2.7 Observability

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- **NFR-24:** JSON logs shall be generated for every API call and AI model latency.
- **NFR-25:** Critical system errors shall trigger alerts (via slack or email) within **5 minutes** of detection.

2.8 Privacy

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- **NFR-26:** The system shall comply with **Egyptian Data Protection Law**.
- **NFR-27:** Minimal personal data shall be stored, and identifiers shall be hashed or pseudonymized.
- **NFR-28:** Businesses shall be able to export or delete their stored data.

Actor	Description
End User (Customer)	A visitor or client interacting with the business chatbot via widget or Meta integration.
Business Admin	Registered business staff managing chatbot configurations, viewing analytics, and editing the knowledge base.
AI System	Orchestrator responsible for intent detection, emotion analysis, RAG retrieval, and response generation.
Backend API (FastAPI)	Middleware managing requests between user, AI models, and the database (PostgreSQL).
Database (Neon PostgreSQL + pgvector)	Stores all persistent entities: users, sessions, messages, knowledge base, and analytics reports.